BTS(C) - V - (S) – 05 – 042 (G)

B.Tech. Degree V Semester (Special Supplementary) Examination July 2005

ME 505 HEAT AND MASS TRANSFER

(2002 Admissions)

Time: 3 Hours — Maximum Marks: 100

(Data Book may be permitted)

I a) Define thermal conductivity, thermal resistance and thermal conductance. What is the approximate range of thermal conductivity of solids, liquids and gases? (10)

b) Saturated steam at 110°C flows inside a copper pipe (thermal conductivity 450W/m.k) having an internal diameter of 10cm and an external diameter of 12 cm. The surface resistance on the steam side is 12000 W/m^2 and that on the outside surface of pipe is 18 W/m^{2}K. Determine the heat loss from the pipe if it is located in space at 25°C. How this heat loss would be affected if the pipe is lagged with 5cm thick insulation of thermal conductivity 0.22W/mk? (10)

OR

II a) Explain a method to obtain solutions to the problem of transient heat conduction in infinitely thick solids. (10)

b) Determine the time required to heat a steel plate of 24mm thick from the initial temperature of 25°C to final temperature of 450°C by placing the plate into a furnace at 600°C.
Take the following properties of steel
K=45.4 W/m-°C, C_p= 0.5 KJ/Kg-°C, $\rho = 7800\, Kg/m^3$
h(heat transfer coefficient between the plate and surroundings in the furnace) = 23.3W/m^2 °C. (10)

III a) Differentiate between mechanisms of heat transfer by free and forced convection. Mention some of the area where these mechanisms are predominant. (10)

b) Determine the heat loss per hour from a wall of a building when the wind is blowing parallel to its surface with a speed of 2km/hour. The wall is 5 metre long and 3 metre high. The temperature of the wall is 25°C and air temperature is 5°C. (10)

OR

IV a) What do you understand by the hydrodynamic and thermal boundary layers? Illustrate with reference to flow over a flat heated plate. (10)

b) A square plate 40cm x 40cm maintained at 400°k is suspended vertically in atmospheric air at 300°k:
(i) Determine the boundary layer thickness at trailing edge of the plate
(ii) Calculate the average heat transfer coefficient using a relation N_u=0.516 $(G_L\, P_R)^{0.25}$ (10)

V a) Define a geometrical or shape factor and what are its salient features. (10)

b) Determine heat lost by radiation per meter length of 80mm diameter pipe at 300°c, if
(i) located in a large room with red brick walls at a temperature of 27°C
(ii) enclosed in 160 mm diameter red brick conduit at a temperature of 27°C.
Take $\in$ (pipe) = 0.79 and
$\in$ (brick conduit) = 0.93 (10)

OR

(Turn Over)

VI a) Derive a general relation for the radiation shape factor in case of radiation between two surfaces. (10)

b) Determine the radiation heat flux between two closely spaced, black parallel plates radiating only to each other if their temperatures are 850k and 425k respectively. Calculate the heat flux presuming that each of the parallel plates has an emissivity of 0.5. In each case, the plates have an area of $4m^2$. (10)

VII a) (i) Discuss the importance of heat exchangers for industrial use.
(ii) What are the basics for the classification of heat exchangers
(iii) Sketch a two-shell pass, four tube pass, reversed current heat exchanger. Label the different parts. (10)

b) Air enters a cooler at 145°c and at 3 ata and is brought to 45°C by passing through tubes of 10 mm ID surrounded by water which enters the cooler at 15°C and leaves at 30°C. Assuming the heat exchanger is counter flow, find the mean temperature difference.

If the air velocity in the tube is limited to 6.5 m/sec., find the length of the tube required –

Neglect the tube resistance and assume water side heat transfer Coefficient is $435KJ/m^2$-hr-°c. (10)

OR

VIII a) Derive the relationship between the effectiveness and the number of transfer units for a counter flow heat exchanger. (10)

b) Calculate the outside tube area for a single pass steam condenser to handle 35000kg of dry and saturated steam at 50°c. The tubes are 25mm outside diameter and 22.5 mm inside diameter and a tube material has a thermal conductivity of 90kcal/m-hr-°C. The average water velocity in each tube is limited to 2m/sec.
Assume steam side film coefficient is 16500 KJ/m^2-hr-°C and inlet and outlet temperatures of water are 15°C and 25°C. (10)

IX a) (i) Define the process of mass transfer and list some industrial applications where mass transfer is involved.
(ii) Distinguish between molecular diffusion, thermal diffusion, pressure diffusion and forced diffusion. (10)

b) Estimate the diffusion coefficient of carbon tetrachloride into air from the following data recorded in a Stefan-tube experiment with Carbon tetrachloride and Oxygen:
Diameter of tube and its length above liquid surface: 1 cm and 15cm respectively.
Temperature and pressure maintained: 0°C and 760 mm of mercury.
Evaporation of Carbon tetrachloride : 0.03gm in 10 hour
Vapour pressure of Carbon tetrachloride: 33mm of mercury (10)

OR

X a) Water evaporator from the surface of a Lake. Derive Stefan's equation for the rate of evaporation. State the assumptions made. (10)

b) Water is available at the bottom of a well which is 2.5 m in diameter and 5m deep. Estimate its diffusion rate into dry atmospheric air at 25°C. The diffusion coefficient for the system is approximated to be 0.0925 m^2/hr and the atmospheric pressure is $10N/cm^2$. (10)

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